

Math 152 Lecture Notes: Week 2

Miles Kretschmer

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0.1 More words on ε and δ

0.1.1 An example

Let $f(x) = \sqrt{x}$. We will first show that this is continuous at any $a > 0$. That is, for any $\varepsilon > 0$, there is a $\delta > 0$ such that if $|x - a| < \delta$, then $|\sqrt{x} - \sqrt{a}| < \varepsilon$. This time we use the factorization

$$|\sqrt{x} - \sqrt{a}| \cdot |\sqrt{x} + \sqrt{a}| = |x - a|$$

and so

$$|\sqrt{x} - \sqrt{a}| = \frac{|x - a|}{|\sqrt{x} + \sqrt{a}|} < \frac{|x - a|}{\sqrt{a}}$$

provided that $x \geq 0$ so $\sqrt{x}$ is well defined. Thus if we want to ensure that $|\sqrt{x} - \sqrt{a}| < \varepsilon$, we need $|x - a| < |a|$, so $x \geq 0$, and $|x - a| < \sqrt{a} \cdot \varepsilon$, so that

$$\frac{|x - a|}{\sqrt{a}} < \varepsilon$$

To make both of these things true, we can take $\delta = \min(a, \sqrt{a} \cdot \varepsilon)$ works.

Let's check that this works. If $|x - a| < \min(a, \sqrt{a} \cdot \varepsilon)$, then $|x - a| < a$, so $x \geq 0$ and $\sqrt{x}$ is well defined. Also, $|x - a| < \sqrt{a} \cdot \varepsilon$, and so

$$|\sqrt{x} - \sqrt{a}| = \frac{|x - a|}{|\sqrt{x} + \sqrt{a}|} < \frac{|x - a|}{\sqrt{a}} < \frac{\sqrt{a} \cdot \varepsilon}{\sqrt{a}} < \varepsilon$$

as required.

The function $f(x) = \sqrt{x}$ is also continuous at 0, in the sense that $\lim_{x \rightarrow 0^+} f(x) = 0 = f(0)$ (the function is not defined for $x < 0$, so the right limit is all we can do.) Indeed, given any $\varepsilon > 0$, if $x > 0$ and $x < \varepsilon^2$, then $|\sqrt{x}| < \varepsilon$. Thus we can take $\delta = \varepsilon^2$ in this case.

0.1.2 What I expect you to be able to do with limits

First of all, why do we define the limit this way, in terms of ε 's and δ 's? The answer is that there is really no other precise way to define the limit. I will use the precise definitions of limits and continuity this week to explain why the fundamental theorem of calculus is true. After taking care of these foundational matters, this class will not focus much on evaluating or verifying limits, but I do want you to understand what the definition means and how it works.

As for checking the ε, δ definition of continuity for particular functions, there is no universal method for doing this, different functions require different ideas. I expect you to be able to do this for simple functions like linear functions and quadratic polynomials which we have seen already. I will also expect you to know that certain functions are continuous, including polynomials, the square root function, and sine and cosine functions, and to use the properties of limits and continuity to verify the continuity of functions obtained by combining these. You should also be able to evaluate limits of rational functions that come out to 0/0 by factoring the numerator and denominator, and apply the squeeze theorem.

0.2 Plan for this week

Our main goal this week is to understand the fundamental theorem of calculus. We will define the integral of a continuous function, and see how this definition is related to area. We will then discuss what the fundamental theorem of calculus asserts and why it is true. This theorem will give us the most effective way to compute integrals, and we will start building up a repertoire of computational techniques.

1 Definite Integrals

You may have some familiarity with integrals already, but we will start from the foundations and build up to the point of computing integrals.

We start by considering an instance of integration which has a straightforward interpretation, the definite integral of a continuous function f which is nonnegative, $f(x) \geq 0$ for $x \in [a, b]$. This is denoted

$$\int_a^b f(x)dx$$

and represents the area A between the x -axis $y = 0$ and the graph of f . [Draw picture.] As with the limit before, we will translate this geometric description into a definition in terms of the values of f . Our approach to defining the integral will be to put upper and lower bounds on this area, in such a way that we can make these bounds as close together as we want. We then define the integral to be the unique number below every upper bound and above every lower bound.

In order to approximate the integral from above and below, we divide the interval $[a, b]$ into smaller subintervals, and estimate the area above each subinterval using rectangles. Specifically, we choose points $x_0 < x_1 < \dots < x_n$ in $[a, b]$, with $a = x_0$ and $x_n = b$, and consider the intervals $[x_{i-1}, x_i]$ for $i = 1, 2, \dots, n$. Such a division of the interval into subintervals is called a partition

$$\mathcal{P} = \{a = x_0 < x_1 < \dots < x_{n-1} < x_n = b\}$$

On each interval in the partition $[x_{i-1}, x_i]$, f attains a minimum $f(x_{\min}) = \ell_i$ and a maximum $f(x_{\max}) = u_i$, by the Extreme Value Theorem.

The area of the region above each $[x_{i-1}, x_i]$ is bounded below by $\ell_i(x_i - x_{i-1})$ and bounded above by $u_i(x_i - x_{i-1})$. [Draw picture]. Letting $\Delta x_i = x_i - x_{i-1}$, we have the following estimates on the area A .

$$L(f, \mathcal{P}) = \sum_{i=1}^n \ell_i \Delta x_i \leq A \leq \sum_{i=1}^n u_i \Delta x_i = U(f, \mathcal{P})$$

The expressions labeled $L(f, \mathcal{P})$ and $U(f, \mathcal{P})$ are called lower and upper sums, respectively.

To define the integral, we consider all possible partitions of the interval.

Proposition. *Provided that f is continuous, then one can choose a partition that makes the upper and lower bounds as close together as we want. That is, for any $\varepsilon > 0$, there is a partition $\mathcal{P}$ such that $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$.*

Thus considering all the upper and lower bounds, we converge on a unique value for the area. [Draw picture with lots of lower and upper bounds converging on a value.]

Definition. The integral of f over $[a, b]$,

$$\int_a^b f(x)dx$$

is the unique number above every lower bound $L(f, \mathcal{P})$ and below every upper bound $U(f, \mathcal{P})$.

Notice that this definition makes sense for any continuous function, even those which take negative values. The geometric interpretation changes however. For a general continuous f ,

$$\int_a^b f(x)dx$$

represents the area above the x -axis minus the area below the x -axis. [Draw picture with colors.]

Example 1. Let $f(x) = c$ be constant. Then for any partition $\mathcal{P}$ of $[a, b]$, we have $\ell_i = u_i = c$ for all i , and so

$$U(f, \mathcal{P}) = L(f, \mathcal{P}) = \sum_i c \cdot \Delta x_i = c \sum_i \Delta x_i = c(b - a)$$

Therefore,

$$\int_a^b c dx = c(b - a)$$

as expected. [Draw picture]

Remark. It is not really tractable to apply this definition to compute integrals of all but the most simple functions. In the book, you can see how to evaluate integrals of x and x^2 using this definition. This week we will learn a much more effective way to compute integrals. It is, however, important that the integral is defined like this. This definition comes from the interpretation of the integral as an area in the case we are considering, and underlies other interpretations of the integral in geometry, physics, and beyond.

Later, we will consider another interpretation of this definition of the integral: recovering displacement from instantaneous velocity. This will be a prototype of the fundamental theorem of calculus, which relates differentiation and integration and provides the most effective way to compute integrals.

Remark. One does not need to consider the absolute maximum and absolute minimum to get a good approximation of the integral. For any choice of sample point $t_i \in [x_{i-1}, x_i]$, we have

$$L(f, \mathcal{P}) \leq \sum_{i=1}^n f(t_i) \Delta x_i \leq U(f, \mathcal{P})$$

so if $L(f, \mathcal{P})$ and $U(f, \mathcal{P})$ are close together (and so close to the true value of the integral) the sum in the middle will also be close to the true value. In fact, one can show that for any $\varepsilon > 0$, for any sufficiently fine partition and any choice of t_i , the sum above is within ε of the true value of the integral. This provides another way of defining the integral, which is often useful in proving theorems about integration.

2 First Properties of the Integral

Working directly with the definition, we can check some basic properties of the integral which will be important later. The first lets us put upper and lower bounds on the integral of a function by comparing it to other functions.

Proposition. Suppose that f and g are continuous functions on an interval $[a, b]$, and that $f(x) \leq g(x)$ for all $x \in [a, b]$. We then have the same inequality between their integrals:

$$\int_a^b f(x)dx \leq \int_a^b g(x)dx$$

In particular, if $c_1 \leq f \leq c_2$ for constants c_1 and c_2 , we have the following estimates:

$$c_1(b-a) \leq \int_a^b f(x)dx \leq c_2(b-a)$$

Proof Sketch. If $f \leq g$ on $[a, b]$, then for any partition $\mathcal{P}$ we have $L(f, \mathcal{P}) \leq L(g, \mathcal{P})$ and $U(f, \mathcal{P}) \leq U(g, \mathcal{P})$. Considering all partitions, this implies that

$$\int_a^b f(x)dx \leq \int_a^b g(x)dx$$

□

The second property allows us to decompose the interval into two subintervals, and break up the integral as a sum.

Proposition. Suppose $a \leq b \leq c$, and f is continuous on $[a, c]$. Then

$$\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$$

Sketch. It suffices to show that the sum of the two integrals is above every lower bound and below every upper bound on the integral over $[a, c]$. Given any partition $\mathcal{P}$ of $[a, c]$, we can refine it by adding b as a point to obtain $\mathcal{P}'$. Replacing a partition with a finer one can only make the lower bound larger and the upper bound smaller [draw picture]. The upper and lower sums for $\mathcal{P}'$ can be split into upper and lower sums for the intervals $[a, b]$, and $[b, c]$, from which we obtain

$$L(f, \mathcal{P}) \leq L(f, \mathcal{P}') \leq \int_a^b f(x)dx + \int_b^c f(x)dx \leq U(f, \mathcal{P}') \leq U(f, \mathcal{P})$$

□

It will be useful to make this more flexible, so that we do not need to worry about the order of a, b, c . So far, we have only defined

$$\int_a^b f(x)dx$$

when $a \leq b$. For $b < a$, we define

$$\int_a^b f(x)dx := - \int_b^a f(x)dx$$

Then the following statement reduces to the previous one.

Proposition. If $f(x)$ is continuous on an interval containing a, b, c , then

$$\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$$

regardless of the order of a, b, c .

Remark. For integrals where $b < a$, the inequalities in our estimates on the integral get reversed, because everything is multiplied by -1 .

3 The Fundamental Theorem of Calculus

To motivate the fundamental theorem of calculus, we will consider another interpretation of the integral: recovering the displacement of an object given its velocity at each time. At this point the discussion will be informal, it will be made more formal later.

[Draw picture]

Consider an object moving along a line, with velocity $v(t)$ at time t . We will assume that v is a continuous function of t . How can we estimate the position $x(t)$ of the object given this information? Well, on any interval of time $[a, b]$, the function v attains a minimum v_ℓ and a maximum v_u . (Here I allow negative velocities, they mean that the object is moving left. The object should be to the right of where it would be if it had velocity v_ℓ the entire time, and to the left of where it would be if it had velocity v_u the entire time. That is

$$v_\ell \cdot (b - a) \leq x(b) - x(a) \leq v_u(b - a)$$

If you are always going between 3 and 5 miles per hour for an hour, you will have traveled between 3 and 5 miles. We can get a more precise estimate for the displacement $x(b) - x(a)$ by dividing up the interval into pieces, estimating the displacement on each one, and adding them up. These estimates are precisely the lower and upper sums $L(v, \mathcal{P})$ and $U(v, \mathcal{P})$. We expect then that the displacement is the integral of the velocity:

$$x(b) - x(a) = \int_a^b v(t) dt$$

On the other hand, velocity is the rate of change of displacement, $v(t) = x'(t)$. This suggests that differentiation and integration are inverse operations, an observation which is made precise in the following theorem.

Theorem 1 (Fundamental Theorem of Calculus). *Let f be a continuous function on some interval I .*

1. *For $a \in I$, function*

$$F(x) = \int_a^x f(x) dx$$

is differentiable on I with derivative $F'(x) = f(x)$.

2. *For $a, b \in I$, if G is any function with $G' = f$, then*

$$\int_a^b f(x) dx = G(b) - G(a)$$

The conclusion (2) will be the most important for us, as it provides the most effective way to compute integrals. On the other hand, most of the conceptual weight of the theorem is in (1): (2) is a fairly straightforward consequence of (1).

Proof of (1). We need to show that for any fixed x ,

$$\lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = f(x)$$

The expression $F(x+h) - F(x)$ can be written

$$\int_a^{x+h} f(t) dt - \int_a^x f(t) dt = \int_x^{x+h} f(t) dt$$

Thus, we need to show that

$$\lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t) dt = f(x)$$

We can check the left and right limits separately. I do the right limit, the left is essentially the same after dealing with some negative signs. We will check this limit using the ε, δ definition. The idea is that for h small, f is close to $f(x)$ on $[x, x+h]$ because f is continuous, and so the integral above is close to $h \cdot f(x)$. Consider $\varepsilon > 0$. As f is continuous at x , there is a δ such that if $|y - x| < \delta$, then $|f(y) - f(x)| < \varepsilon$. We will show that if $0 < h < \delta$, then

$$\left| \frac{1}{h} \int_x^{x+h} f(t) dt - f(x) \right| \leq \varepsilon$$

(In the definition of limit, it does not matter whether we use a strict or nonstrict inequality as we can always replace ε by $\varepsilon/2$.) So assume $0 < h < \delta$. Then, on the interval $[x, x+h]$, we have the estimates

$$f(x) - \varepsilon < f < f(x) + \varepsilon$$

[Draw picture] This in turn gives us the estimates

$$h \cdot (f(x) - \varepsilon) \leq \int_x^{x+h} f(t) dt \leq h \cdot (f(x) + \varepsilon)$$

Dividing through by h (remember $h > 0$ here) we get

$$f(x) - \varepsilon \leq \frac{1}{h} \int_x^{x+h} f(t) dt \leq f(x) + \varepsilon$$

This is exactly what we needed to show. □

Now to explain why (2) is true, we introduce some language. A function G with $G'(x) = f(x)$ is called a *antiderivative* of f . Part (1) tells us that

$$F(x) = \int_a^x f(t) dt$$

is an antiderivative of f . On the other hand, if F and G are both antiderivatives of f , then

$$\frac{d}{dx}(G(x) - F(x)) = f(x) - f(x) = 0$$

We now invoke the following fact:

Proposition. *If $H'(x) = 0$ on an interval, then H is constant on that interval.*

This means that $G(x) - F(x)$ is a constant C , and so $G(x) = F(x) + C$. Taking F to be the antiderivative we defined in (1), we have

$$G(b) - G(a) = F(b) + C - (F(a) + C) = F(b) - F(a) = \int_a^b f(t) dt$$

This proves (2) in the fundamental theorem of calculus.

4 Some Integral Calculations and Properties of the Integral

Now that we have the fundamental theorem of calculus, we can use it and our knowledge of derivatives to evaluate some integrals and get some general properties of the integral.

Example 2. We will show that

$$\int_a^b x dx = \frac{b^2 - a^2}{2}$$

In particular for $a = 0$, we get the familiar expression for the area of a triangle. [Picture] This follows from the fundamental theorem (2) as $x^2/2$ is an antiderivative of x .

More generally, for $n \neq -1$ we have

$$\int_a^b x^n dx = \frac{b^{n+1} - a^{n+1}}{n+1}$$

This applies to negative and fractional powers as well

$$\int_a^b x^{p/q} dx = \frac{b^{p/q+1} - a^{p/q+1}}{p/q+1}$$

although we now need to worry about where $x^{p/q}$ is continuous, so we should probably take $0 < a, b$ to be safe.

Example 3. What is

$$\int_4^9 \sqrt{x} dx?$$

Using $\sqrt{x} = x^{1/2}$, we get

$$\frac{9^{3/2} - 4^{3/2}}{3/2} = \frac{2}{3} (27 - 8) = \frac{2 \cdot 19}{3} = \frac{38}{3}$$

Imagine how difficult this would be to justify using upper and lower sums!

Now that we know how to integrate powers, we can integrate polynomials using the following.

Proposition (Linearity of the Integral). *Suppose f and g are continuous functions on $[a, b]$ and c is a constant.*

1.

$$\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

2.

$$\int_a^b c \cdot f(x) dx = c \int_a^b f(x) dx$$

Proof Sketch. Let F and G be antiderivatives of f and g respectively. Then $F + G$ is an antiderivative of $f + g$, and $c \cdot F$ is an antiderivative of $c \cdot f$. Using the fundamental theorem of calculus to evaluate the integrals using these antiderivatives, we have (1) and (2). $\square$

Example 4. What is

$$\int_1^2 (x^3 - 3x + 1)dx?$$

This breaks up as

$$\int_1^2 x^3 dx - 3 \int_1^2 x dx + \int_1^2 1 dx = \frac{2^4 - 1^4}{4} - 3 \frac{2^2 - 1^2}{2} + (2 - 1) = 15/4 - 3/2 + 1 = 17/4$$

5 Indefinite Integrals

When we evaluate integrals using the fundamental theorem of calculus, the main problem is to find an antiderivative of our function f . We will introduce some language and notation to talk about this in short order. First though, we need to consider the following question:

Question. Suppose we know that the velocity of an object has velocity $v(t) = 1 - t^2$ at time t . Is this enough information to know the object's position at time $t = 1$?

Answer. No. The point is we also need to know the object's position at some point in time. If we do not know this, then $x(t) = t - t^3/3$ and $x(t) = t - t^3/3 + 4$ are both possibilities for the object's position, so we do not know what $x(1)$ is. Mathematically, the point is that antiderivatives are not unique. $\square$

On the other hand, any two antiderivatives differ by a constant. That is, if F is some antiderivative of f , then any other antiderivative has the form $F + C$ for some constant C . We saw this earlier in our discussion of the fundamental theorem of calculus. This means that if we know the value of the antiderivative at some point, we can solve for the constant and get a unique function.

Question. Suppose we now additionally that $x(0) = 1$. Now can we find the position at $t = 1$?

Answer. Yes. We know $x(t) = t - t^3/3 + C$ for some constant C . Plugging in $t = 0$, we have $1 = x(0) = 0 - 0 + C$, and so $C = 1$, $x(t) = t - t^3/3 + 1$. Plugging in $t = 1$, we get $x(1) = 1 - 1/3 + 1 = 5/3$. $\square$

Given a function f , we write

$$\int f(x)dx$$

to denote an arbitrary antiderivative of f . This expression is called an *indefinite integral* as opposed to a definite integral, which is a number. Because antiderivatives are not unique, this is not a single well defined function. When we evaluate an indefinite integral, we always include an arbitrary constant in our formula, like so:

$$\int x dx = \frac{x^2}{2} + C$$

This means that any antiderivative of x has the form $x^2/2 + C$ for some choice of constant C .

Remark. The dx in this notation is a bookkeeping device which will be useful when we use changes of variables to compute integrals.

With this device, computing a definite integral

$$\int_a^b f(x)dx$$

has two steps:

1. Evaluate the indefinite integral

$$\int f(x)dx$$

to get a description of all antiderivatives of f . Fixing a value of the constant, we get a particular antiderivative F .

2. Evaluate $F(b) - F(a)$.

The second step is straightforward, so most of the time we will be interested in the first step, evaluating the indefinite integral. We can write down what we were doing before in this language.

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \text{ for } n \neq -1$$

We also have the following linearity properties:

Proposition.

$$\int (f(x) + g(x))dx = \int f(x)dx + \int g(x)dx$$

$$\int c \cdot f(x)dx = c \int f(x)dx$$

Some remarks are in order on how to interpret these, as both sides are only defined up to a choice of arbitrary constant. For example, we write

$$\int x dx = x^2/2 + C_1$$

$$\int x^3 dx = x^4/4 + C_2$$

and so

$$\int (x + x^3)dx = x^2/2 + x^4/4 + C_1 + C_2?$$

Well, the $+C_i$ in each case just stands for an arbitrary constant, and a sum of two constants is a constant, so we can just write

$$\int (x + x^3)dx = x^2/2 + x^4/4 + C$$

In general, you can consolidate all arbitrary constants into a single one. The precise statement here is that if F is an antiderivative of f and G is an antiderivative of g , then $F + G$ is an antiderivative of $f + g$, exactly what we saw before.

Using our knowledge of derivatives, we can also put together a list of basic indefinite integrals:

$$\int a dx = ax + C \text{ for } a \text{ constant}$$

$$\int x^{p/q} dx = \frac{x^{p/q+1}}{p/q+1} + C \text{ for } p/q \neq -1$$

$$\int \cos(x) dx = \sin(x) + C$$

$$\int \sin(x) dx = -\cos(x) + C$$

$$\int \frac{1}{\cos(x)^2} dx = \tan(x) + C$$

Example 5. Let's evaluate the following indefinite integral:

$$\int (x^3 - 1 + 2 \cos(x)) dx$$

Using linearity, this breaks up as

$$\int x^3 dx - \int 1 dx + 2 \int \cos(x) dx$$

This comes out to

$$x^4/4 - x + 2 \sin(x) + C$$

consolidating all the constants as before.

For more general functions built up using products, quotients, and composition, there is no straightforward algorithm to evaluate indefinite integrals as there is for derivatives. This is no accident, integrals of straightforward looking algebraic functions can give us new transcendental functions. We can, however, evaluate lots of integrals using what we have now. Indeed, every rule for differentiation can be read backwards to give a rule for integration. We will start learning these methods of integration next week.

When evaluating *definite integrals*

$$\int_a^b f(x) dx$$

we can use the following notation to track what we are doing. First, evaluate the indefinite integral and choose a particular antiderivative F :

$$\int f(x) dx = F(x) + C$$

Then the definite integral is expressed as

$$\int_a^b f(x) dx = F(x)|_a^b = F(b) - F(a)$$

5.1 Questions for Group Work

Question. What is the area between the graph of $y = \sin(x)$ and the x -axis between $x = 0$ and the next point where the graph crosses 0? [Draw picture]

Question. Evaluate

$$\int_1^5 \frac{3+x}{\sqrt{x}} dx$$